

Multiple Choice

1. **Answer:** D

Options: A) $f[n] = 4 nu[n] + [(5/6)n - 1] 6^n u[n]$; B) $f[n] = 6 nu[n] + [(2/3)n - 1] 3^n u[n]$; C) $f[n] = 3^{(n+1)}u[n] - 5^n u[n]$; D) $f[n] = 3 nu[n] + [(4/5)n - 1] 5^n u[n]$; E) $f[n] = [(2/3)n + 1] (3/2)^n u[n] - 2^{(n-1)} u[n]$

Note: Correct option: D - $f[n] = 3 nu[n] + [(4/5)n - 1] 5^n u[n]$

2. **Answer:** E

Options: A) the output is independent of the clock; B) the output is constantly changing; C) the output is triggered by a rising edge clock; D) the output is LOW; E) the output follow input

Note: Correct option: E - the output follow input

3. **Answer:** A

Options: A) 0.119 wavelength; B) 0.130 wavelength; C) 0.112 *wavelength*²; D) 0.100 wavelength; E) 0.123 *wavelength*²

Note: Correct option: A - 0.119 wavelength

4. **Answer:** A

Options: A) 45,000 lb; B) 18,000 lb; C) 28,000 lb; D) 30,000 lb; E) 15,000 lb

Note: Correct option: A - 45,000 lb

5. **Answer:** A

Options: A) $1000 \sin 10^6 t$; B) $100 \sin 10^6 t$; C) $50 \cos 10^6 t$; D) $1000 \cos 10^6 t$; E) $0.1 \cos 10^6 t$

Note: Correct option: A - $1000 \sin 10^6 t$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.7 meters, 1 meter'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.30'.

9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '1/3'.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 176%. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 100 *meter*². Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key